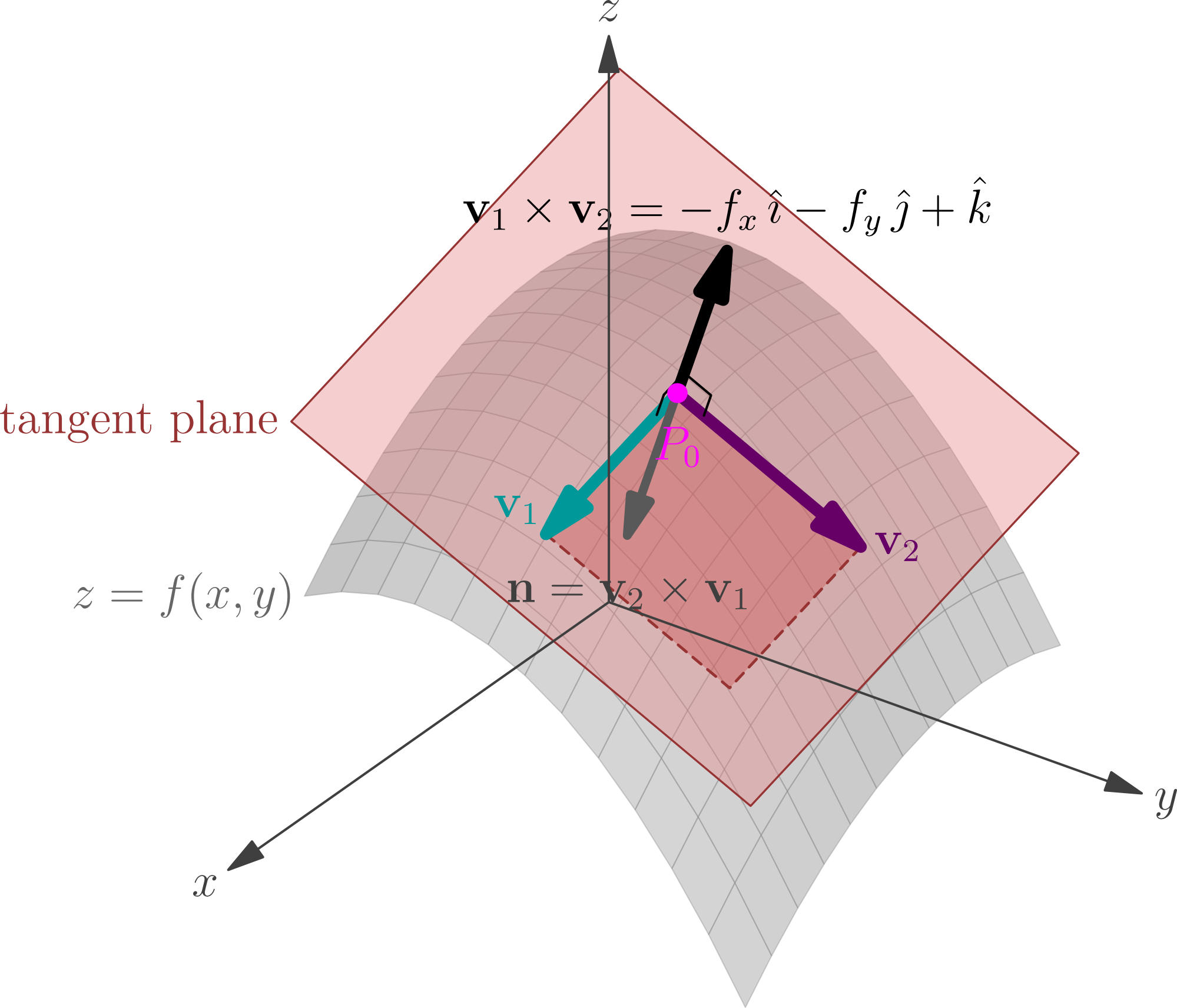
z
$\mathbf{v}_1 \times \mathbf{v}_2 = -f_x\,\hat{\imath} - f_y\,\hat{\jmath} + \hat{k}$
tangent plane
P_0
$\mathbf{v}_1$
$\mathbf{v}_2$
$z = f(x, y)$
$\mathbf{n} = \mathbf{v}_2 \times \mathbf{v}_1$
x
y